A Review of Rotor Wake Physics and Modeling







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This paper presents an overview of developments in rotorcraft wake modeling, particularly in the past decade, from the experimental, theoretical, and computational viewpoints. Current understanding of wake physics and limitations in wake modeling are summarized and juxtaposed with trends experimental and numerical research. Advances in flow imaging and velocimetry have provided evidence that supports simplified models of rotor wake behavior. Persistence of tip vortices to long ages, deterministic rollup phenomena leading to the transition to the far wake, and entrainment of a part of the nascent tip vortex (TV) vorticity into the counterrotating inboard wake have been shown through experiments. Advances in computational fluid dynamics have improved near- and mid-wake modeling capabilities. Hybrid methods that employ vortex element, vorticity transport, or vorticity confinement methods will be necessary for long-age (far-wake) modeling in the near future. A new hover experiment with high-accuracy measurements that tie the blade loading, rotor performance, and wake characteristics is one suggestion to facilitate numerical model correlation and development.

Nomenclature

A_{mr}	area of the main rotor disk, m^2
C_{Peng}	power coefficient
C_W	weight coefficient, $C_W = \frac{W}{\rho A_{\text{mr}} V_t^2}$
c	local blade chord, m
e	energy, $\text{kg-m}^2/\text{s}^2$
k	multiplication factor
M	Mach number
n	number of mesh points
R	rotor blade radius, m
r	local radial blade location, m
u, v, w	velocity in a Cartesian coordinate axis, m/s
V_t	tip speed, m/s
W	weight, N
x, y, z	distance in a Cartesian coordinate axis, m
α	angle of attack, deg
ρ	density, kg/m^3
ω	vorticity magnitude, s^{-1}
∞	freestream

Introduction

Many of the issues facing rotorcraft researchers today intricately involve the rotor wake. Operation in urban environments requires accurate control of the wake to minimize noise due to blade–vortex interaction (BVI) and rotor–wake interaction with nearby infrastructure. The need to understand and resolve brownout, in which the rotor wake entrains sand or other particulates and convects them into the pilot's visual field, has become critical for safety in desert environments. Sling load dynamics and shipboard operations require further understanding of the interaction of the wake with its environment. Strong interest in rotor wakes comes from the wind turbine community, driven by the need for knowledge on convection and dissipation of individual wakes in wind farms.

The rotor wake can be deconstructed into three major "fields": near, mid, and far, as illustrated in Fig. 1. The near field includes the region where the wake initially leaves the rotor blade and completes formation of the tip vortices. The accurate solution of this region is important to the prediction of blade airloads, BVIs, rotor vibration, and aeroacoustic signatures. The midfield encompasses the helicopter fuselage. Wake resolution is important to characterize rotor–fuselage interactions (RFIs), interior noise, and empennage buffeting. The far field includes the region where wake–environment interactions such as ground effect, brownout, shipboard operations, and formation flying are important.

This paper summarizes the state of knowledge and advances with respect to rotor wakes from both experimental and computational perspectives, building on previous surveys of the field. A starting point is the review of helicopter aerodynamics done by Phillippe et al. (Ref. 1) in 1985, as part of an AGARD (Advisory Group for Aerospace Research and Development) volume on Aeromechanics. A more recent perspective comes from Conlisk (Ref. 2) published in 1997, the same year that the American Helicopter Society conducted a Technical Specialists' Meeting on rotorcraft aeromechanics and acoustics. A history of computational developments for rotorcraft by Strawn (Ref. 3) in 2006 provides a recent perspective. In March 2009, the U.S. Army Research Office sponsored a workshop at the Georgia Institute of Technology (Ref. 4). This paper

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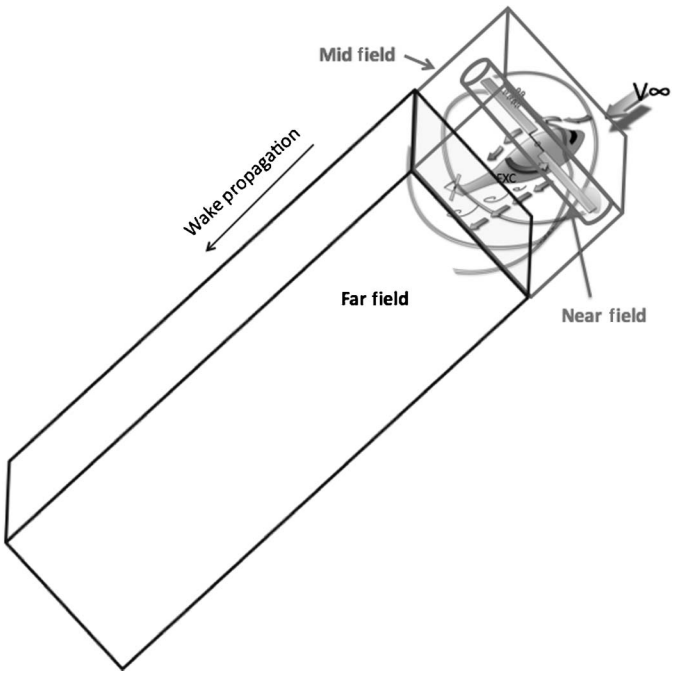


Fig. 1. Schematic of the near-, mid- and far-field helicopter wake regions. Permission for use of the helicopter cartoon by F.X. Caradonna, AFDD.

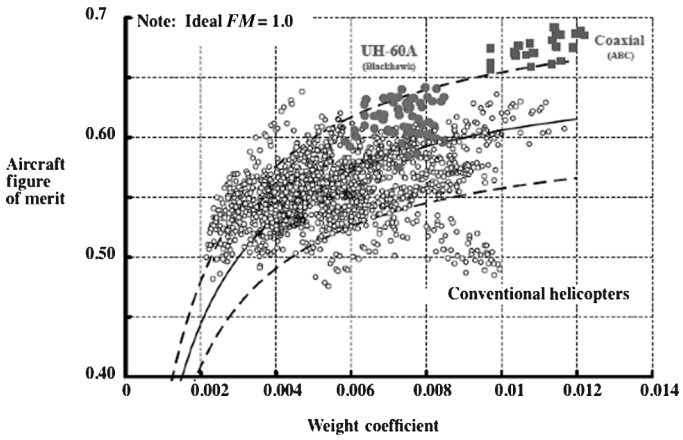


Fig. 2. Improving rotor figure of merit at higher thrust coefficients requires improved rotor wake prediction capability. Originally from Harris (Ref. 120); this modified version from Tung (Ref. 121).

summarizes the technical discussions from the workshop, in the context of experimental and predictive milestones, particularly during the past decade. In each section, a discussion of the state-of-the-art and recent advancements is first presented, followed by a discussion of the pertinent workshop questions with regard to each perspective. Finally, a list of conclusions and recommendations extracted from the workshop is provided.

Figure 2 illustrates the advancement, and the need for further advancement, in helicopter performance prediction from 1985 to the present. The rotor figure of merit (FM) has risen considerably at larger weight (thrust) requirements, and along with it, the need for accurate prediction has increased. In 1985, prediction of blade aerodynamics was primarily accomplished via panel and blade element-based potential flow methods. Navier–Stokes calculation of a rotor in hover was just being demonstrated, but it was still far too slow and expensive to be used as industry tool. Advances in computing power and algorithms have now rendered Reynolds-averaged Navier–Stokes (RANS) calculations viable even for use some design stages of rotorcraft. The issues in blade computational fidelity have shifted to turbulence modeling for the wake and separated rotational flows.

Meanwhile, new strides are being made in characterizing the wake and related flow field experimentally and with analytically based computational tools. Application of accurate nonintrusive measurement techniques in facilities, where vortex–wall interactions are minimized, have enabled clean measurements of vortex core structure in axial and in edgewise flight to large wake ages. With clear evidence from such experiments, analytical tools have progressed to open the way toward simulation of flight conditions where wake behavior at long ages is important.

A recent U.S. Army Research Office (ARO) workshop, held in March 2009, posed eight questions about the status of wake modeling to researchers interested in rotorcraft and wind turbine wakes:

1) What is the structure of the TV at its origin, and what factors influence it?

2) How much of the blade bound vorticity actually ends up in the TV?

3) How fast does the TV diffuse/dissipate/decay?

4) What is the role of turbulence in these processes?

5) What are the physical phenomena responsible for observed "vortex jitter"?

6) What is the state of computational capability for the TV as a function of age in hover and forward flight?

7) How well are long-age phenomena, such as ground vortex rollup and tail rotor/main rotor interaction, captured?

8) What is the status of turbulence modeling for helicopter rotor wakes?

Experimental Advancements

The wake of a rotor is at once an extremely complex flow field, and one where the dominant phenomena are amenable to simple description. While by no means exhaustively including all of the research that has been performed, Table 1 classifies available experimental databases and Table 2 lists several of the issues that have been addressed during these experimental campaigns. The Gray deconstruction (Refs. 5, 6) of the hover wake vorticity structure into tip vortices and helical vortex sheets of the 1950s, and the Landgrebe extension (Ref. 7) into the complex interactions of forward flight developed in the early 1960s, still hold true to a very large extent. Several other details and implications have been proven since then. Many of the early results were obtained by

Table 1. Organization of experimental test cases

Experiment Type	Reference
Reviews	Desopper (Ref. 125)
Hover	Gray (Ref. 5,126), Castles (Ref. 127), Landgrebe (Ref. 128), Caradonna and Tung (Ref. 17), Tung et al. (Ref. 16), Piziali and Felker (Ref. 129), Lorber (Ref. 130), and Norman and Light (Ref. 131)
Axial	Castles (Ref. 127) and Caradonna et al. (Ref. 132)
Edgewise	Gray (Ref. 6), Wilson and Mineck (Ref. 133), Landgrebe et al. (Ref. 134), and Lorber (Ref. 135)
Ground effect	Empey (Ref. 136), Curtiss (Ref. 137), Light (Ref. 138), Cimbala (Ref. 139), and Brand (Ref. 25)
Brownout	Johnson and Leishman (Ref. 140)

Table 2. Issues explored in experiments

Type of Analysis	References
Model scale, Reynolds number, Mach number	Hein and Chopra (Ref. 141)
2D versus 3D nonrotating versus rotating	McAlister (Ref. 142)
Inflow	Elliott and Althoff (Ref. 143) and Peters (Ref. 144)
Blade–vortex interaction	Hubbard and Harris (Ref. 145), Lorber (Ref. 146), Caradonna (Ref. 132), and HART-II Team (Ref. 115)
Stall	McCroskey and Fischer (Ref. 147), Bousman (Ref. 148), Carr (Ref. 149), Chandrasekhara and Carr (Ref. 150), and Yu (Ref. 151)
Near wake	McAlister (Ref. 152), Thompson (Ref. 153), Wadcock (Ref. 10), Light (Ref. 154), Felker (Ref. 155), and Adams (Ref. 156)
Transition/pairing	Caradonna et al. (Ref. 122)
Circulation/vortex strength	Thompson (Ref. 153), Ghee (Ref. 12), and Kim (Ref. 11)
Turbulence issues	Ramasamy et al. (Refs. 21, 22), Mahalingam and Komerath (Ref. 157)

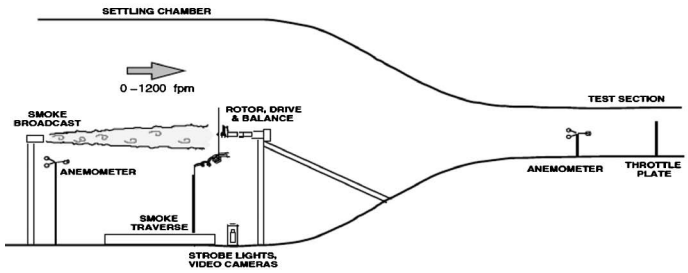
careful analysis of photographs and motion pictures taken of wakes with entrained light-scattering particles fed into the rotor inflow, or through the tip. such as titanium dioxide formed by the reaction of $TiCl_4$ fed from blade tips, with air. However, these techniques had limited three-dimensional depth resolution. In the 1980s, it became possible to obtain excellent spatial resolution, accuracy and precision using laser sheets to illuminate thin, well-defined slices of the flow field. Mechanical chopper wheels synchronized to the rotor, Bragg cell beam choppers, and pulsed lasers enabled excellent temporal resolution as well, progressing, respectively, from time-averaged to purely simple harmonic phenomena to more complex events involving shear layer interactions. The advent of laser Doppler velocimetry (LDV) complemented the capabilities of mechanical probes, such as hot-wire and hot-film anemometers, to obtain instantaneous velocity, resolved by direction, at accurately specified points in the inflow and wake, noting, however, that accurate velocity measurement inside the core of the TV core still posed substantial difficulties. In the 1990s, light sheets made from machine-vision strobes were used to mimic pulsed laser sheets and applied to rotor flow fields in large facilities, at realistic Reynolds numbers. In the past decade, particle image velocimetry (PIV) and its stereoscopic version have enabled capture of two- and three-component velocity fields, giving a more complete picture of the wake at various conditions; the difficulty of seeding the flow with particles small enough to follow the sharply accelerated flow inside the TV core, yet large enough to be captured in PIV, persist. Findings on several issues are summarized below.

Findings on wake features

The transition to the far wake occurs through deterministic, periodic, repeatable pairing events, for a two-bladed rotor. The transition to the far wake has been shown to occur through deterministic vortex-pairing rather than through chaotic processes, even at high Reynolds number, for a two-bladed rotor in axial flight or hover. Conducting clean hover experiments in ground facilities has always proved to be difficult, because of unsteadiness induced by the interaction of the wake with facility walls and fixtures—itself a manifestation of the long persistence of tip vortices. Figure 3 depicts an experimental setup used at Ames Research Center



(a) Test section looking from upstream



(b) Schematic sketch of the setup and light sheet orientation

Fig. 3. Facility used to conduct interference-free axial flow experiment (Ref. 122) for a two-bladed rotor in the Ames 30 × 30 settling chamber of the 7 × 10 tunnel.

in 1996 by Frank Caradonna and colleagues to study the performance of a rotor in axial flow. They explored the idea of obtaining hover performance as the asymptotic limit of zero climb rate. The rotor is mounted along the axis of the $7' \times 10'$ wind tunnel in the $30' \times 30'$ settling chamber rather than in the test section. As a part of the experimental effort, a set of white light sheets and intensified cameras captured the behavior of the wake of this rotor. The tip Mach number of the rotor was near 0.7, and the Reynolds number was in the typical operating regime of a tail rotor. Figure 4 shows a sequence of video images at 9° collective and 3.5 fps "climb speed," clearly showing how tip vortices (numbered for identification) remained as strong entities beyond 720° of vortex age, and then interacted and merged during the transition to the far wake. Until this merger occurred, the vortex trajectories remained cleanly periodic and repeatable. Core growth appeared to be less than 50% at 720° of wake age. However, when the tunnel flow velocity was reduced below a certain point, where the settling chamber velocity fell below 1 m/s, the entire wake became unstable. This was attributable to the fact that vortices encountered downstream obstructions and meandered back upstream, causing violent transient interactions. This hypothesis was reinforced when obstructions were placed downstream, and it was observed that the behavior of the wake could be forced to become unsteady even at higher velocities. For higher climb speeds, the nature of the test configuration with the long return path and large settling chamber minimized the possibility of vortices interacting with obstructions downstream and returning to the inflow plane in a way that is usually difficult to achieve with ground-based "hover chambers." Thus, this evidence also indicates that vortices persist as distinct entities far beyond the traditional

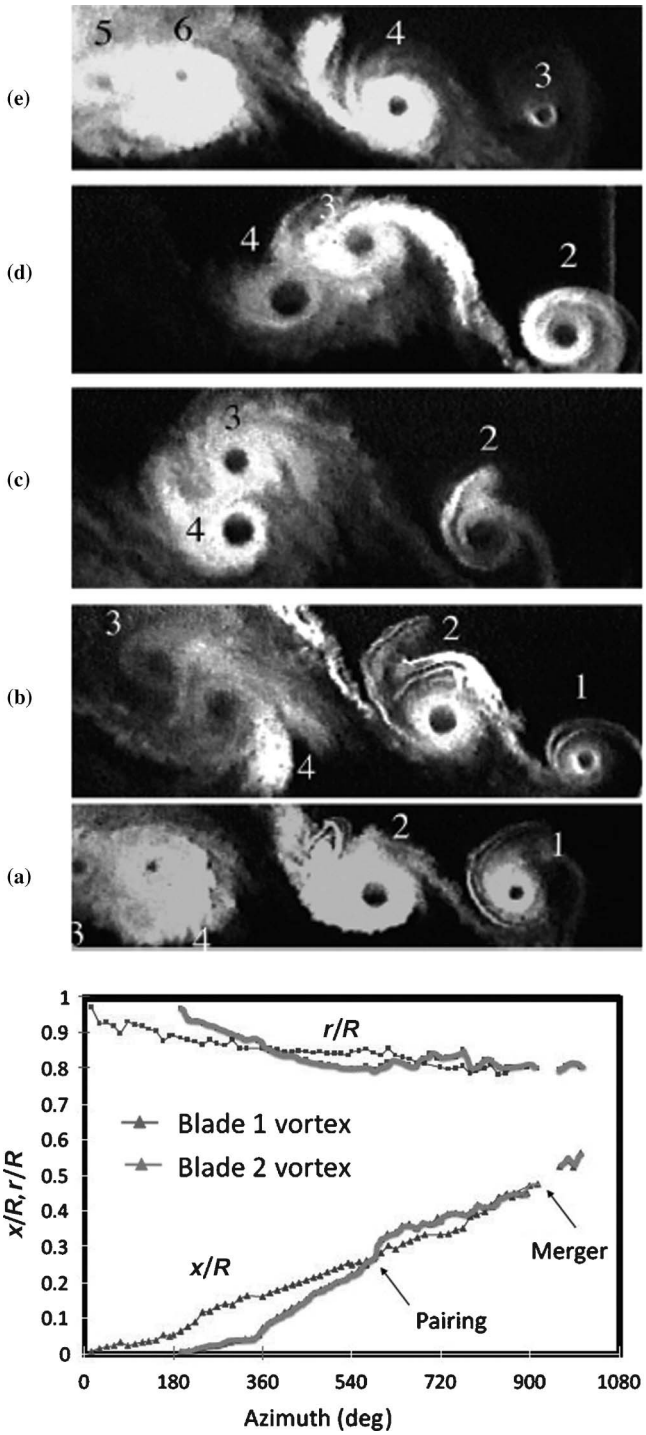


Fig. 4. Vortex pairing sequence in the wake of the two–bladed rotor at 9° collective, 3.5 fps axial climb speed, in the facility shown in Fig. 3 (Ref. 122).

age limits, where they were believed to enter the far-field region and dissipate.

Similarly, in the case of a rotor with one or two blades, unexplained "jitter" phenomena have mostly been shown to be adequately explained through predictable vortex interactions, whether in the near or far field of the rotor. The evolution of the near wake thus in fact conforms quite well to the schematic descriptions developed by Gray for the near wake in hover and Landgrebe for the wake in forward flight, except that the mutual vortex interactions included by Landgrebe also occur in the hover case as the wake ages. The tip vortices roll around each other, in behavior that is similar in appearance at any given cross-section to the interaction of two corotating vortices in a two-dimensional shear layer. These are strongly helical vortices, however, and thus the stretching associated with this rollup must have strong influences. The rollup proceeds until the cores come close together and then lose definition in the merger. It is logical to expect turbulent processes to become significant at this stage, and that the strong axial flow slows down, with an accompanying rapid growth in core size. The merged vortices may then weaken much quicker than their components did before merger, and this delineates the start of the "far wake." The discrete structures in the far wake may then break down, as occurs in shear layer growth. While these processes look very complex as the number of blades increases, they still obey deterministic descriptions until the vortex mergers occur.

It is not known whether this deterministic, periodic behavior can be repeated as the number of blades is increased above the two blades in the experiment, and specifically if the number of blades goes above four. Researchers on vortex interactions, related to shear layer transition, have reported mathematical treatments indicating that a system with four or more vortices in proximity will develop chaotic behavior. In the rotor case, the vortices are not completely independent, so it may well be that the progression to chaos if any, does not occur quickly enough to influence the deterministic pairing interactions and the transition to the "far wake" condition. In cases such as the onset of the vortex ring state or the formation of the ground vortex and the development of "brownout" in dusty environments, several turns of the tip vortices may indeed remain close enough for chaotic behavior to be significant. Such phenomena remain to be investigated. However, the visible proof of deterministic behavior for a two-bladed rotor encouraged Jain and Conlisk (Ref. 8) to undertake calculations of wake vortex behavior to long wake age and to study mutual interactions of the vortices. They were able to capture the mutual interaction of the vortices observed in the 1996 Ames experiments and the resulting vortex trajectory. This capability allowed them to obtain better agreement with other published experimental wake trajectories than was possible from prior lifting line/free wake calculations. This also allowed computation of vortex rollup and reingestion phenomena in ground effect. Thus a simple model of deterministic vortex interactions, based on observed physics, has resulted in both efficient and accurate prediction of rotor wakes to large ages.

Vortex core size remains small for large vortex ages, up to several revolutions. Further proof of the long persistence of tip vortices comes from visualization of the density gradient across TV cores. Figure 5 is a NASA Ames shadowgraphy experiment (Ref. 9) on a Sikorsky S-76 rotor at a tip Mach number of 0.605. Thus the highest Mach number difference across the core can only be 0.6, even if the TV starts out at a full strength value computed using potential flow theory from the peak bound circulation at the tip. It is well known that shadowgraphy and schlieren techniques do not work (i.e., do not provide enough contrast to be captured in video) for differences below Mach 0.25. In other words, even after some 360° of rotor age, the vortex strength remains close to its original strength.

Coalescing the above two observations, one may conclude that the tip vortices in hover or low-speed climb should stay well organized in deterministic rather than chaotic behavior, and their strength should dissipate very slowly with increasing vortex age, at least in the case of rotors with up to two blades. Figure 6, which summarizes data from several independent experiments, clearly illustrates this behavior. The factor k is used to multiply the peak bound circulation to match the vortex strength data. This appears to be in contradiction to some existing formulations, where a large turbulent diffusion is necessary to explain

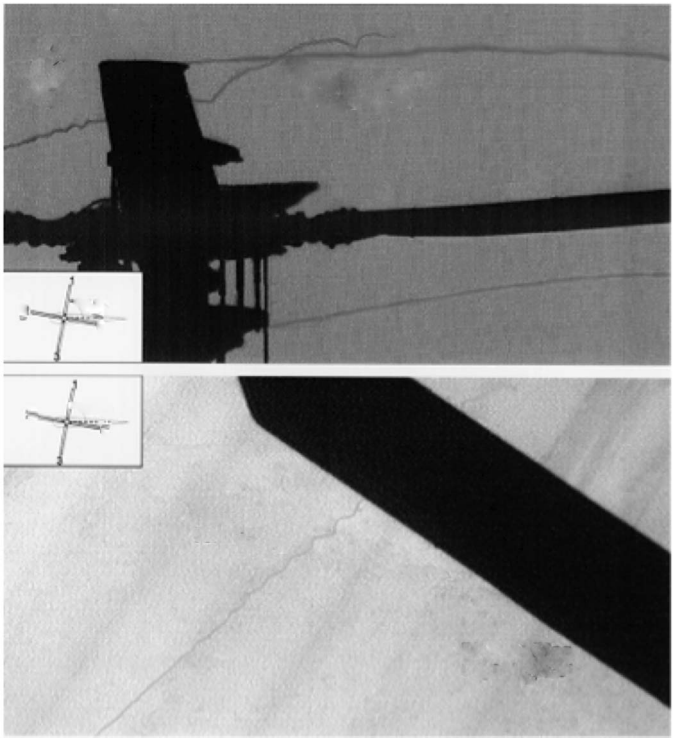


Fig. 5. Shadowgraphy visualizes the tip vortices from an S-76 rotor, at a tip Mach number of 0.605 (Ref. 9).

why the measured TV strength is much lower than the peak bound circulation on the blades.

One significant experimental observation remains, countering the above conclusion about deterministic pairing and periodic behavior. The aforementioned shadowgraph experiments showed some curious phenomena, indicating a sharply helical local motion of the vortex filament itself, with very short wavelength. This could be interpreted as an offset and precession of the core from the nominal trajectory of the TV. Curiously, this does not appear to lead to immediate breakdown of the TV, as might have been expected in the case of the wing tip vortices from commercial airliners. Figure 5 also illustrates other phenomena that may be described as "jitter" of the vortices. These may be due to instabilities triggered by blade–vortex or vortex–vortex interaction, and it may lead under some circumstances to an obstruction of the axial flow in the vortex core. Obstruction of the axial flow is accompanied by swift growth and destruction of the vortex core. Wadcock (Ref. 10) has reported a sharp contraction in TV core size that he attributes to a strain on the vortex due to interaction with the first blade passage. He notes that earlier reports of rapid core expansion and diffusion can be traced to the use of only a single-point sensor across a vortex under facility test conditions that exhibited cycle to cycle "jitter."

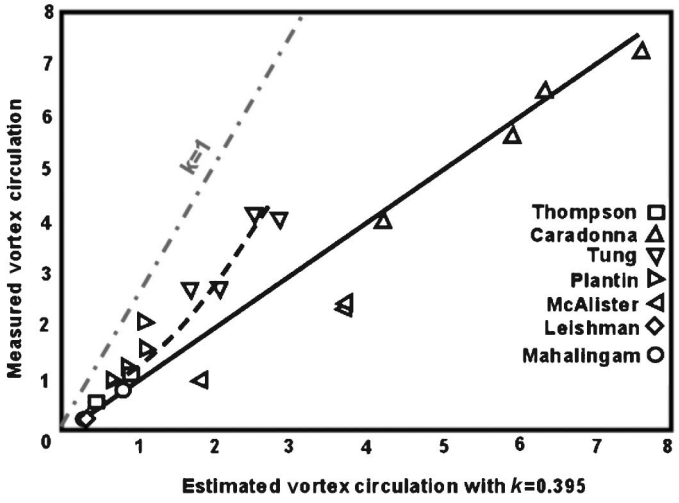
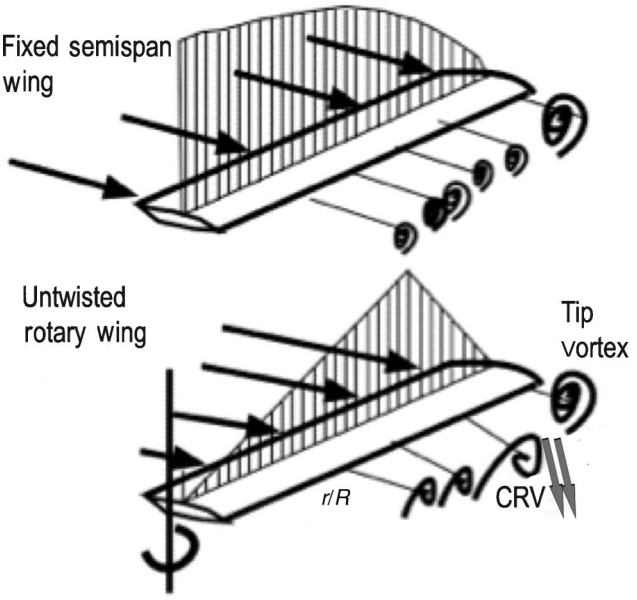


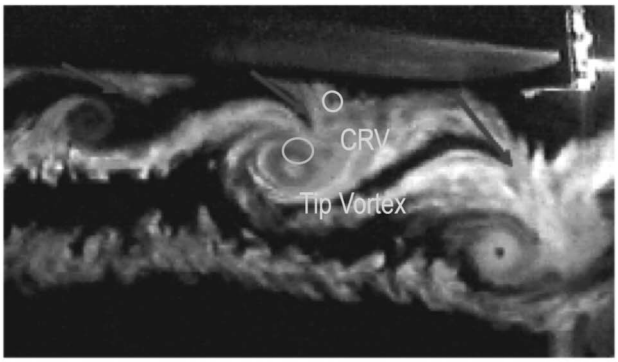
Fig. 6. Measured tip vortex strength compared with the estimated value of peak-bound circulation near the tip of the corresponding rotor blades, from experiments in various facilities (Ref. 123).

The rolled up structure at the edge of the inboard vortex sheet. As the tip vortices develop along helical trajectories, the inboard blade wake, which is basically a thin, flat shear layer developed from the turbulent boundary layers over the surfaces of the blade, convects down faster than the TV. Interactions with older TV segments may occur. Kim et al. (Ref. 11) demonstrated that in the case of a highly loaded, untwisted blade, the edge of the inboard vortex sheet of the rotor blade rolls up into a counterrotating vortex (CRV), whose strength approached 50% of that of the TV. Experiments by Ghee and Elliott (Ref. 12) at Langley employing a substantially larger (but still model scale) rotor confirmed this finding. While this has not been directly shown in numerical computations, Egolf (Ref. 13) cites calculations that attribute up to a 20% reduction in TV strength to the interaction with the inboard vortex sheet. He also demonstrated by smoke visualization images from UTRC wind tunnel tests and computations that a "dual vortex" or counterrotating vortex pair can be seen above the rotor blade in some flight conditions. The experimental cases of Kim and Ghee established that such a structure must exist in many model-scale experiments at moderate or higher thrust coefficients. This finding will have a profound influence on the relation between the peak-bound circulation on the blade and the strength of the TV. If this presence of a strong rolled up CRV exists for full-scale main rotors at high-thrust coefficients, it should have substantial effects on the structure of the wake. Hence it should also have a strong effect on induced velocities and the thrust that the rotor develops. However, there is no evidence of similar counterrotating vorticity in the computations reported by Narducci (Ref. 14) for a large fixed blade twisted to approximate the circulation distribution of a typical rotor in hover. These computations were correlated to experiments at the Boeing Vertol Wind Tunnel for a vortex shed from a nonrotating generic tiltrotor blade at $M_\infty = 0.15$ and $\alpha = 6.5^\circ$. The circulation distribution shown in this case had a small hump near the tip, but this does not appear to generate a visible counterrotating vortex structure at the edge of the inboard sheet. One may postulate that the counterrotating pair phenomenon is due to aspects of the rotation, such as radial acceleration, that are not present in fixed-wing experiments.

The vortex strength paradox. Early applications of the "force-free wake" analysis resulted in a disconnect between the correct force-free wake geometry calculated from the blade loading for known thrust, and the correct thrust calculated using a prescribed (known) wake geometry. Recent results suggest that this can be attributed to a basic misconception in the application of potential flow vortex element (VE) analysis to rotor wakes. Patient application of nonintrusive measurement techniques in facilities where vortex–wall interactions are minimized has enabled clean measurements of vortex core structure in axial and in edgewise flight to large wake ages. This has spurred a retrospective evaluation of data existing in the literature. The traditional assumption that all the vorticity outboard of the bound circulation peak will roll into the TV appears to be too simplistic. A large part of this vorticity is convected into the edge of the inboard vortex sheet. The extent to which this occurs



(a) Schematic illustration



(b) Laser sheet image of the vortex pair for a two-bladed rotor

Fig. 7. Tip vortex behavior for fixed and rotating blades in forward flight (Refs. 11, 123).

depends on the details of the blade tip shape, demanding high-resolution numerical techniques to capture in predictive analysis.

Mahalingam (Ref. 15) correlated the highest TV strength reported in measurements by several researchers, including his own LDV data on a two-bladed rotor in forward flight, with the estimated peak bound circulation near the tips of the corresponding rotor blades. Figure 6 from his thesis indicates that up to 55% of the peak-bound circulation may be lost from the TV shortly downstream of the blade tip. This is true as well for the Carradonna–Tung two-bladed rotor test case that is often used as a validation data set for computations. The outlier datum in Fig. 6, showing over 90% capture of the tip-bound circulation peak is from an experiment conducted by Tung and Caradonna (Ref. 16). This experiment was a follow-up to the Caradonna–Tung test case. It appears that the low TV strength, corresponding to only about 50% of peak-bound circulation obtained in the original set (Carradonna–Tung) puzzled the researchers. Consequently, they modified their curve-fitting procedure to capture the vortex strength by extending the interpolation region further away from the location of TV core passage and used the induced velocity over this wider region to estimate the TV strength. As illustrated in Fig. 7, the near-saw-tooth shape of the bound circulation on a rotor blade, as opposed to the monotonic shape on a corresponding fixed wing, implies that the sense of rotation of the inboard vortex sheet is indeed counter to that of the TV. Where the inboard sheet rolls into a CRV, the counterrotating vortex pair induces a strong downward flow. A measurement technique that detects only the induced velocity magnitude cannot distinguish the effect of the TV from that of the CRV. Thus if the induced velocity is used as a measure of TV strength, and the measurement point is in between the TV and CRV, then a spuriously high estimate of the TV strength will result. However, this estimate may come closer to the value of the peak-bound circulation. This appears to have occurred in the case of the outlier point of Fig. 6. Thus it appears that the low estimate of the initial paper by Caradonna and Tung (Ref. 17) was the correct one. Wong (Ref. 18) used LDV to capture the detailed evolution of the vortex structure over the blade tips of the two-bladed rotor used by Mahalingam, and showed that the outer part of the shear layer rolling up over the tip would be entrained into the CRV, not the TV. As a result, the fully formed TV would only contain part of the vorticity in the shear layers coming off the blade tip. Vasilescu (Ref. 19) showed that the rollup of the TV over a rounded blade tip was substantially different from that over a squared-off blade tip.

In the absence of such a realization that not all of the blade tip circulation will roll into the TV, the discrepancy between the TV strength and the peak-bound circulation has caused much uncertainty. The role of turbulence to explain this behavior has generated diverging hypotheses. Some researchers assumed that the TV must have an initial strength equal in magnitude to the peak bound circulation, and hence attributed the reduction within about 45° of vortex age to turbulent diffusion of vorticity. Jain and Conlisk (Ref. 8) have shown through order of magnitude estimates that such a loss could not be due to turbulence, and indeed if there were any turbulent phenomena that caused such a sharp drop in vortex strength, there is no way to explain the long persistence of the vortex shown, for instance, in Figs. 4 and 5. Instead, the presence of turbulence should diffuse or dissipate the vortex swiftly, within 90° of vortex age. This conundrum led to much confusion in the 1990s. In several experiments, researchers used single-point laser velocimeter probes to scan across vortex cores, using azimuth-resolved sampling synchronized to the rotor. When the vortex position "jittered" due to the transient vortex interactions from cycle to cycle discussed previously, the effect was that the apparent vortex core was "smeared" over several degrees of rotor azimuth, losing definition, and leading to erroneous agreement with CFD codes that suffered from large numerical dissipation due to poor grid resolution and low-order spatial algorithms. Adding to the uncertainty were theoretical analyses, coming at a time when "chaos theory" was seen as the avenue to predict turbulence, suggesting that the wake of a rotor must be fundamentally unstable, swiftly degenerating into chaos.

With the introduction of PIV, this error should be avoidable, although the resolution of the core velocity profile from PIV is difficult because the core rarely has sufficiently detectable seed particles in it, and those that are seen inside may not follow the local flow direction due to centrifugal effects. This problem is exacerbated as the rotor scale and Reynolds number increase, the vortex gets stronger, and measurement distances increase. Recently a series of papers (Refs. 20–22) reports on experiments using stereoscopic PIV that indicate that the turbulence in the shear layers rolls up into the vortex core. Based on these experiments, a Reynolds number–based vortex model has been proposed to empirically simulate this behavior (Ref. 23). These tests note "significant turbulence activity up to two core radii from the tip vortex axis." Asymmetric flow characteristics in the tip vortices were associated with "a pronounced anisotropic distribution of eddy viscosity, a typical characteristic of flows with high streamline curvature."

The sudden descent problem. Brand (Refs. 24,25) has analyzed the flight experiences of the sudden descent problem encountered by tilt rotor

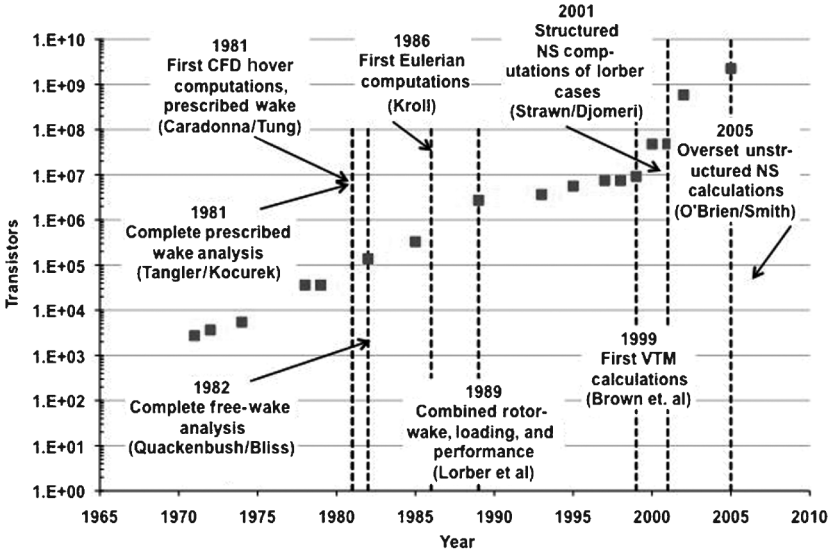


Fig. 8. Microprocessor development predicted by Gordon Moore (Ref. 28) compared with numerical simulations for hovering rotors.

aircraft. He shows that the catastrophic loss of lift suspected in this problem is actually a case of the rotor blade experiencing sharp downflow, resulting in reduction of angle of attack, rather than any possibility of rotor stall. The downflow is a result of the accumulation of tip vorticity into the "vortex ring state." There is some similarity in the fluid mechanics regarding the accumulation of tip vortices into a strong ground vortex in ground effect. The established vortex ring was shown to be a self-propelling structure that can lead or follow the descending rotor. An increase in advance ratio is seen to be the way to escape this condition.

Wake interactions in the far field. A recent source of knowledge on the behavior of rotor wakes is the surge in research on wind turbines. The Reynolds number range of interest here is even larger than that involved in full-scale helicopter aerodynamics, but the tip Mach numbers are generally low. Two samples of current work in this field were reported at the Workshop. Massouh et al. (Ref. 26) report detailed PIV data in the wake of wind turbine models in a low-speed wind tunnel, albeit at relatively small model scale and Reynolds number.

Another valuable aspect of wind turbine research to those interested in rotorcraft aerodynamics is that the interaction between the wakes of different, closely spaced turbines are of strong interest, and large-scale visualizations are available on such interactions in wind farms from natural condensation or smoke studies. Dobrev and Massouh (Ref. 27) also showed analyses of the interactions between wakes of turbines. Although recent progress using LES and Eulerian vortex methods has been made, these analyses in most production engineering settings still require more efficient singularity-based methods to minimize the numerical demands posed by the multiple interactions.

The above experimental findings enable a new look at prediction techniques, potentially offering great simplification. Meanwhile, issues such as turbulence continue to be important in predicting the initial rollup of shear layers into tip vortices, and in predicting the size evolution of the TV and the spreading of the inboard blade wake. These, in turn, are essential for future efforts to design blades for low BVI loads, low noise, and better performance.

Computational Advancements

The development of computer hardware and its associated cost continue to follow the 1965 trend predicted by Gordon Moore (Ref. 28), now known as Moore's law. Microprocessor capacity has increased exponentially, whereas cost has decreased similarly, so that coupled with the development in parallel computing capability, computational solutions of rotors in hover have likewise progressed from the solution of potential-based equations of a few thousand nodes to Navier–Stokes-based equations with millions of nodes (Fig. 8).

Development of computational methods

Since the advent of the computer, rotorcraft researchers have looked to computational means as a mechanism to improve their knowledge of the physics of the complex rotor flow field and to improve engineering predictions of rotorcraft behavior for design, simulation, and analysis. A review of the development of primitive-variable computational fluid dynamics (CFD) for rotorcraft published by Strawn (Ref. 3) is recommended as a general review. A synopsis of the history of these methods, specifically with regards to rotorcraft wake predictions is provided here.

Vortex element methods. The earliest line of numerical research in wake simulations has resulted in a set of VE methodologies that are typically implemented in a class of codes known as comprehensive codes. A review of the most popular of these methods has been detailed in Refs. 29 and 30. Here, the rotor wake is modeled as one or several VEs trailing from each blade using the potential or "singularity-based" simplification of the fluid mechanics equations of motions. Earliest models assumed the wake structure to be composed of a TV plus a vortex sheet, based on the Gray's (Ref. 6) 1956 model. The helical vortex assumption of the steady wake was later modified to include the capability to model "free" wake motion that the prescribed helical formulation did not allow. Multiple trailers of VEs, also known as filaments, were introduced to more physically represent the wake distribution of vorticity, including vortices shed from the root. Egolf (Ref. 13) gives an excellent synopsis of VE methods specific to rotorcraft applications.

There are several methods of implementation of these VE methods currently in use. Lagrangian calculations of incompressible VE dynamics using the Biot–Savart law (Ref. 29) or combinations of Eulerian and Lagrangian methods (Ref. 31) are popular VE methods for rotor wake codes. A very efficient implementation is the constant vorticity contour (CVC) method, where the ordering of the wake structure is a function of the wake vorticity distribution (Refs. 32, 33). Vortex core models are

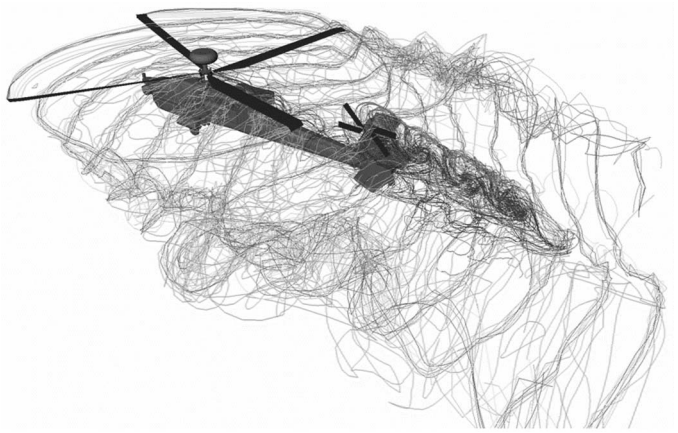


Fig. 9. Simulation of an AH-64 full multirotor configuration using the inviscid Lagrangian wake model, CHARM (Ref. 124).

used to overcome solution instabilities in the near and far wake in these methods. Instability observed in vortex models for hover in the near wake in early studies using time marching schemes appears to have been overcome in more recent work using relaxation schemes.

An alternative implementation of VE methods developed by French researchers (Ref. 34) involves the use of vortex points or blobs to avoid solution instabilities. These methods address issues such as vortex merging, which can be difficult if a specific ordering, such as vortex lattice ordering, is implemented. Conversely, many more points are needed to model the wake, so that the overall cost may increase substantially.

VE methods have been shown to accurately predict the loading and wakes on the rotor in hover and forward flight (Refs. 32, 35–38) for many flight conditions. These methods offer fast turnaround times on a single personal computer (some even real time), but require an estimate of the wake strength and location at the rotor blade. VE methods were first coupled to lifting-line and panel methods, but they have also been coupled to Euler or RANS methods since the advent of RANS as production codes in the 1990s. Because VE methods are free from the numerical diffusion problems afflicting finite-difference or finite-volume CFD schemes (see following section), they are ideally suited to capture the propagation of vortices over long distances and times, as illustrated in Fig. 9. Moreover, vortex methods avoid the complexities of mesh generation and thus are very well suited when frames in relative motion (e.g., rotor blades) need to be modeled.

The cost of the Lagrangian wake calculation scales with the square of the number of elements, so that additional VEs added to the model to refine the wake representation, can result in dramatic computational cost increases. Multiple wake trailers and point vortices can be distributed across a number of processors with varying success. The most successful methods to reduce cost are multipole formulations, such as the fast multipole method (FMM) in wake code known as CHARM (Ref. 33), which reduce the $\mathcal{O}(n^2)$ cost to $\mathcal{O}(nlog(n))$ cost. Computational cost reduction efforts today are focused on CFD and computational structural dynamics (CFD/CSD) coupling trim using VE methods using massively parallel computers.

Potential-based formulations. One early focus of computational efforts was on hybrid methods, where methods to solve the simplified forms of the Navier–Stokes equations for a rotor blade were combined with an analytical wake representation to reduce the size of the problem so that it could be resolved on the limited memory computers that were the norm at that time. Initial simulations of a rotor in hover were performed by Caradonna and Tung (Ref. 17) for a limited computational domain modified by surface transpiration and outer boundary variables determined from an integral solution of the wake outside of the computational boundaries. Comparisons with experimental hover data showed good correlation and spurred further computational endeavors in the field. By the 1980s full potential codes, such as FPR (Ref. 39) were applied to hover. Applying the transpiration concept from earlier successes, they were able to simulate lifting forward flight configurations (Ref. 40) using a finite-difference transonic small disturbance method.

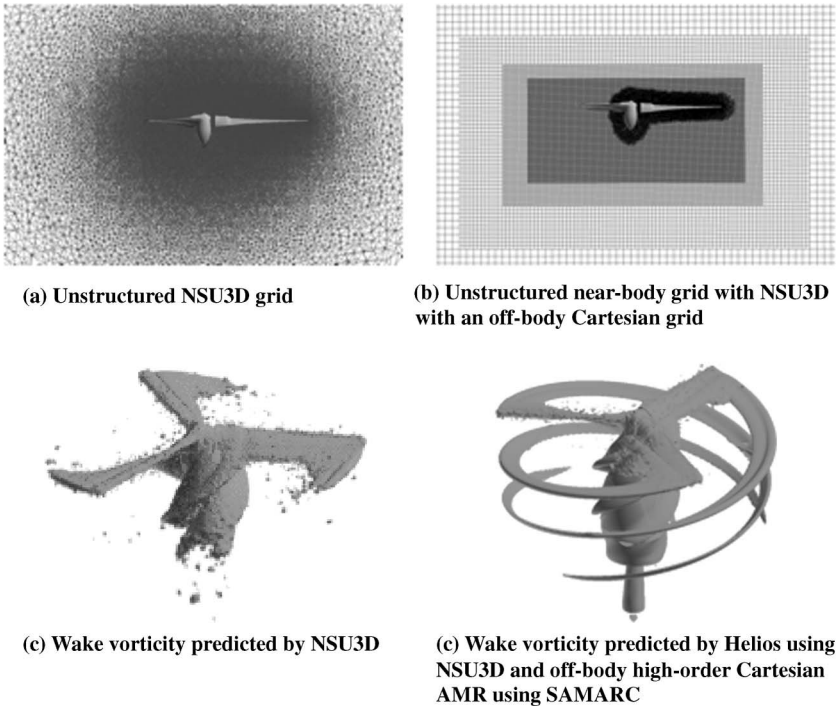
Efforts to develop potential-based formulations of the simplified Navier–Stokes equations slowed during the late 1980s and in the 1990s as computer power increased and Euler/Navier–Stokes methods were developing. One method that continued development during this time was the HELIX hover code (Refs. 41, 42), which is a combined Eulerian/Lagrangian method where the velocity field from the wake vorticity is inserted into the near-blade computations. Recent upgrades to this code replace the potential near-body solver with a Navier–Stokes solver (Refs. 38, 43).

Recently, D'Andrea has revisited the use of panel methods and shown (Refs. 44, 45) that for some flight conditions a full-span unstructured-hybrid panel method is capable of predicting rotor wakes very accurately when coupled to a CVC formulation to capture the wake. He notes that even with a parallel implementation, the use of vortex lattice methods in the wake is impractical for problems such as brownout where 100 rotor revolutions may be necessary (Ref. 46) to capture the in-ground-effect phenomena.

Euler/Navier–Stokes formulations. As noted by Strawn (Ref. 3), one of the major goals of developing Euler/Navier–Stokes methods was the ability to capture rotor wakes without resorting to the hybrid methods necessary with potential-based solvers. Contemporary use of the acronym "CFD" has become synonymous with computational methods that resolve the conservative variable (ρ, ρu, ρv, ρw, ρe) compressible or primitive variable (ρ, u, v, w, e) incompressible formulation of the RANS equations of motion. These methods typically employ finite-difference or finite-volume techniques to resolve the flow field (but not always), and the turbulence closure occurs via an eddy viscosity parameter that is computed from a statistical model of the scales of turbulence. Much of the development of these solvers has focused on the ability to resolve the vortical flow around the blades, so that the TV and vortex sheet can be computed from first principles, ostensibly resulting in a more physically correct simulation.

Conventional rotor configurations are well suited to resolution by overset structural RANS CFD methods, given that the rotor blade planforms can be readily described by quadrilateral elements. For hover cases, a structured mesh composed of a single zone is an efficient way to model the rotor blade. However, CFD methods introduce a numerical dissipation into the solution, so that high-density grids are required to capture the rapidly changing flow field without losing important characteristics. This pertains of course to the topic of interest here, the capture and propagation of the rotor wake. Strawn and Djomehri (Ref. 47) in 2002 computed the wake of a UH-60A rotor in hover using embedded momentum sources to mimic the rotational flow that is generated by the blades of the rotor. They noted a persistent grid-dependence in the simulation. Even with a 64 million node structured grid, the numerical dissipation of vorticity still overwhelmed the physical behavior of their system.

Extending CFD to forward flight means that two frames of motion representing the rotating blades and the stationary background must be computed during the same simulation. Lee (Ref. 48) and Steijl (Ref. 49) have both demonstrated variations of sliding boundary conditions between rotor and background grids, but the most versatile method of



(a) Unstructured NSU3D grid

(b) Unstructured near-body grid with NSU3D with an off-body Cartesian grid

(c) Wake vorticity predicted by NSU3D

(c) Wake vorticity predicted by Helios using NSU3D and off-body high-order Cartesian AMR using SAMARC

Fig. 10. Comparison of wake resolution for TRAM rotor in M = 0.1 climb conditions using NSU3D and Helios (Ref. 104).

modeling different frames of reference is the overset method (Ref. 50). This method provides the ability to model arbitrary motions, including elastic motion, without the problem of overly complex boundary condition development. Overset methods have been successfully implemented and applied in a number of rotorcraft applications (Refs. 51–54).

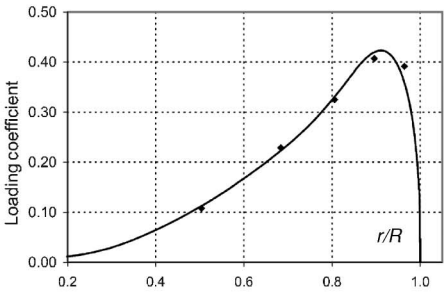
Computational simulation of midfield wakes, such as for RFI, strains even further the resources required by structured methods. Actuator disks will provide good time-averaged loading on fuselages (Refs. 55–57), but the rotor wakes are lost as individual rotor blades are not modeled within the sources. At minimum, time-accurate, blade-referenced source modeling of the rotor is required to obtain an estimate of the instantaneous fuselage surface pressures (Ref. 53) due to the individual blade wake passage. The exact representation of the moving rotor blades is required to evaluate accurately the complete detail of the unsteady wake and its interaction with the fuselage or empennage. Potsdam and Strawn (Ref. 52) studied a full-span V-22 configuration with moving rotor blades. Their structured overset RANS solver required numerous overset grids to model the complex geometry. With a total of 47.6 million nodes, they were able to correlate the unsteady wake–fuselage interaction to adverse flight control and aerodynamic phenomena observed in flight.

Given the limitations of structured RANS methods, unstructured and Cartesian grid topologies have been considered as options to structured grid topologies to reduce the number of computational nodes and the complexity of the grid generation process that is required as a prerequisite for the simulation of the aerodynamics of modern helicopter configurations. Unstructured RANS methods use either a fully tetrahedral grid topology or a mix of prisms, tetrahedral and hexahedral cells to comprise the grid geometry. This approach permits complex geometries to be modeled much more rapidly than is the case with their structured method counterparts, and a single grid surrounding the fuselage can be extended efficiently to form the background or far-field grid. Rapid changes in configurations are also possible with overset methods, permitting hubs, rotors, and struts to be added (or removed) from the model with little difficulty. Unstructured methods are not without their problems, as current solvers cannot efficiently obtain high-order spatial resolution (e.g., fourth or sixth order), and their computational overhead and cost per iteration is significantly higher than with structured methods. Cartesian methods (Ref. 58) do not require body-fitted grids, minimizing the time consumed in grid generation. Problems still exist in modeling rapid geometry changes and in viscous simulations about complex geometries due to the nature of the boundary conditions. A comparison (Ref. 58) of hybrid, Cartesian, and unstructured methods with mixed methods of rotor modeling demonstrated the advantages and shortcomings of these various methods for RFI, whereas Renaud et al. (Ref. 59) compared the performance of two overset structured and one unstructured method with rotor disk models. An alternative that is currently being explored is to use overset near-field grids (structured or unstructured) in conjunction with Cartesian background grids solved by a separate Cartesian solver (Ref. 60), as illustrated in Fig. 10.

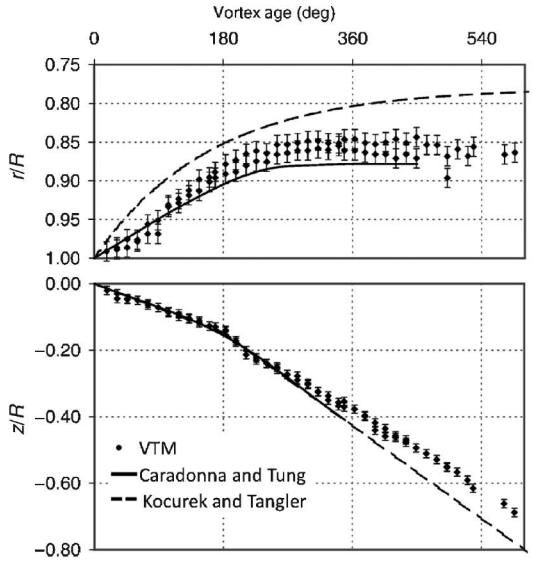
Alternative Navier–Stokes formulations. Traditional RANS CFD methods that solve the variable sets of (ρ, ρu, ρv, ρw, ρe) or (ρ, u, v, w, e) have thus encountered serious problems in propagating the wake without significant dissipation. Research into alternate CFD formulations has sought to alleviate this problem. An alternative approach that has shown great promise to date in addressing long-age wake propagation is the class of methods based on the solution of the RANS equations in vorticity–velocity form. By casting the vorticity as the primary conserved variable, the numerical diffusion of vorticity in the wake is avoided by suitable choice of the numerical algorithm that is used to advect the vorticity through the computational domain. One of the most successful of such CFD approaches is the vorticity transport model (VTM), developed by Brown and his students at Glasgow University (and earlier at Imperial College London) (Refs. 61,62). The VTM method employs a grid-based CFD solver for the wake. This model has been applied to a number of different rotor wake problems with exceptional success to date. These applications include near-field wake modeling for airloads and aeroacoustics (including BVI) (Refs. 63–65), midfield RFI (Refs. 66, 67) and



(a) Wake visualization of a two-bladed hovering rotor (800.000 grid cells, 50 cells/radius, 6 cells/chord)



(b) Blade loding



(c) Comparison of vortex trajectory with experiment

Fig. 11. Hovering wake predictions using the vortex transport method (Ref. 62).

rotor–rotor interaction (Ref. 68), far-field helicopter–aircraft wake interaction (Ref. 69), the vortex ring state (Ref. 70), ground effect (Ref. 71), and brownout (Ref. 46). The CFD solver from the VTM code has been enhanced and implemented in modular form in another code known as VorTran-M with similar successes reported, for example, in wake analyses (Refs. 69, 72).

There has been some effort to solve the vorticity transport (VT) equations by revisiting particle-based methods (Ref. 73) rather than the Eulerian formulation discussed previously. A recent version of the outcome from these efforts is known as the particle vortex transport model (PVTM), although the published record to date (Ref. 74) indicates only partial success. Another implementation (Ref. 75) applies the method, now referred to as the particle vortex method (PVM), to augment a dynamic inflow model within a comprehensive code. While inflow angles are improved, there was no discussion of airloads or of the wake-modeling abilities of the PVM method in the paper. Major hurdles still remain when applying particle methods to high-Reynolds-number three-dimensional flows. If viscous effects are included, then periodic remeshing is needed to maintain accurate representation of the diffusion operators (Ref. 73). For inviscid flows (or the advection step of viscous methods), particle methods suffer from instabilities and loss of accuracy associated with failing to enforce a divergence-free vorticity field and particle disorder (Ref. 73). The apparent flow-modeling flexibility offered by the unstructured and nonconnected particle representation is generally offset by the need in all such methods, to maintain adequate particle overlap for accuracy. Thus when stretching occurs, particles must be continually added to maintain accuracy and limit the nonphysical growth of convection of the vorticity, that is $\mathbf{V} \cdot \Omega$.

A second alternative Navier–Stokes CFD formulation that has shown some success in capturing rotor wakes is the vorticity confinement (VC) method. Developed by Steinhoff (Refs. 76, 77), it has been applied to rotorcraft wakes (Ref. 78) and in particular to brownout (Refs. 79, 80). This method solves the discretized Navier–Stokes equations with the addition of an extra term that confines vorticity to only a few grid cells. As a result, vortices are convected without numerical spreading over long wake ages. The computational savings result from smaller grids, which can be simplified further using Cartesian grids. Qualitatively, the methodology appears to work very well, although detailed correlations to experimental wake data, similar to VTM correlations (Fig. 11), has not, to the authors' knowledge, been published.

All of these methods typically rely on some other computational formulation to calculate the surface loading in the flow so that the vorticity in the near field of the rotor can be resolved. While most of these early efforts have relied on lifting-line or panel-based methods to rapidly generate the surface data, conventional RANS CFD methods still remain as the most accurate for near-blade computations. Thus, a framework

that resolves the alternative RANS CFD methods in the wake while preserving conventional RANS CFD in the rotor near field to take advantage of the best characteristics of both methods is a logical step. These methods are collectively known as hybrid methods and are discussed in the following section.

Hybrid Formulations. As noted in the prior sections, each of the computational methods applied to the wake problem has advantages and disadvantages. It was recognized early on in the 1970s and 1980s that a combination of methods that merge the best characteristics of each method, while mitigating the disadvantages, showed promise in tackling the wake-modeling issue. This class of simulations is known collectively as "hybrid" methods. Early efforts attempted to address the cost of the CFD simulation, along with the problematic wake dissipation, by creating a mixed Eulerian–Lagrangian formulation, although this is no longer the case as hybrid methods that couple traditional CFD with alternative CFD methods such as Eulerian VT formulations are being developed.

Initial hybrid methods typically utilized the more costly computational methods to capture the blade near-field physics, either with full-potential methods in the 1970s and 1980s (Refs. 39, 81–83) or more recently, RANS methods (Refs. 84, 85). These hybrid methods will be referred to as CFD-Lagrangian wake methods. Most of these hybrid methods apply free- or prescribed-wake methods to resolve the far wake, reducing the computational grid requirements. However, they achieve this reduction at the expense of the physics, as the limitations in the free wakes still remain. Additional simplifications, such as modeling one blade of the rotor, further reduce the cost of quasi-steady simulations such as hover and steady level flight.

TURNS-UMARC has shown very good correlation of airloads for forward flight (Ref. 86), and HELIX versions have been shown to predict figure of merit at or within experimental error bounds (Ref. 38). Issues that are still being investigated for these methods tend to center on boundary condition issues associated with the interface when strong levels of vorticity are exchanged between the methods.

More recent hybrid methods that are under development combine the RANS CFD in the blade near field with VT or VC CFD methods in the far field. This hybrid methodology is more consistent than CFD-Lagrangian wake hybrid methods as the Navier–Stokes equations are solved in both the near-field and wake modules. VorTran-M has been coupled with an unstructured solver, RSA3D (Ref. 72), as well as with OVERFLOW and FUN3D, whereas VC has been coupled with TURNS (Ref. 87) and may be coupled with UM-TURNS. Here, the vorticity (or velocity vector) from the wake method can be directly inserted into the CFD grid to alleviate many of the interface issues with which CFD-Lagrangian wake hybrid methods must contend, to conserve the RANS equations. The primary formulations of these vorticity methods still tend to be incompressible, so that density changes must be handled with care to preserve continuity. The current primary drawback with these methods is that their implementation is designed for serial processors, which means that their full effectiveness has not yet been evaluated.

Computational wake modeling issues still to be resolved

While many advancements have been made with computational codes, there are still major wake modeling issues that need to be resolved. Some of the more important questions or observations necessary for wake modeling and the current efforts in the area include the following:

The need for additional research to refine vortex core models. Egolf (Ref. 13) points out that despite large advances in computational methods and computer speed, analytical modeling of rotor wakes continues to be essential due to the high cost of conventional RANS CFD-based methods. Within the VE methods, the core is modeled via analytical algorithms based on empirical observations. Bhagwat and Leishman (Ref. 88) summarize a number of vortex core models and show how the features of the model can impact hover simulations. However, core modeling remains a fertile area for research, as experimental techniques to measure the behavior of the vortex core are improving with technology advancements. These core models were developed from small-scale experiments, so that there are a number of questions relating the model assumptions to the vortex core behavior. Egolf (Ref. 13) specifically cites the assumption of the initial core sizes and decay rates that are necessary when applying VE methods. Both Egolf and Quackenbush (Ref. 33) also list entrainment or agglomeration behavior in the wake for interactional aerodynamics and brownout as an area for additional research.

Expounding on the interactional questions, further understanding of "distinct" vortices (e.g., TV, vortices from abrupt sweep changes), and their interaction with the inboard sheet or with other vortical features of complex blade tip shapes is necessary. For instance, such understanding is essential to correctly model the experimentally observed dual vortices well inboard of the rotor tip, and the rollup of the root vortices.

Near-wake core modeling issues can be alleviated by coupling the VE method with a conventional Euler/RANS CFD code that captures the blade vorticity. Here, the analytical modeling is replaced with first-principles methods, which, if applied correctly, should provide a more physical representation of the near-field wake. However, these methods have their own limitations and are not a panacea for all near-wake issues.

The role of turbulence modeling in defining the near-field (and mid-/far-field) wake. Transition and turbulence modeling are the issues that tend to be most often cited when explaining differences between CFD and experimental results. When resolving the flow field about the rotor blade, the shed wake is directly related to the solver's ability to predict the blade loading and rotor performance. It is this aspect of RANS CFD that has received the bulk of the attention during the CFD development and is summarized very aptly in Strawn et al. (Ref 3). The ability of RANS methods to predict aeroelastic effects will also impact the blade loading and the subsequent strength and location of the prominent wake vortices. Recent CFD/CSD coupling efforts have significantly improved the ability of CFD to predict blade loading (e.g., Refs. 86, 89, 90) and have recently been applied to BVI (Ref. 91)

For rotor applications that include transonic flows and stalled flows, it has been shown that algebraic models do not correctly simulate the behavior of rotor airfoils under these conditions (Ref. 92). One- and two-equation models appear to capture the overall behavior of the rotor well, including the TV, for many applications (Refs. 14, 91). However, they miss dynamic stall rotor characteristics (Refs. 89, 93). Advanced turbulence models are gaining attention after they showed promise in the DARPA Quiet Helicopter program (Ref. 89). Nichols et al. (Ref. 94), Szydlowski and Costes (Ref. 93), and Lynch and Smith (Ref. 95) have recently demonstrated calculations of dynamic stall and improvements in capturing complex wake physics using various advances in turbulence simulation techniques.

Transition remains a key issue in some simulations, in particularly hover, as noted by Narducci (Ref. 14). Many legacy codes assume fully turbulent flows or have some user-defined transition location, but only a few include transition modeling as part of the turbulent closure methodology. Transition modeling appears to have mixed influence on the prediction of loading for dynamic applications (Refs. 96, 97), improving static or low-frequency integrated loads, but having less effect as the reduced frequency increases.

Transition and turbulence modeling issues become even more important when advanced concepts are studied. Active flaps include unsteady

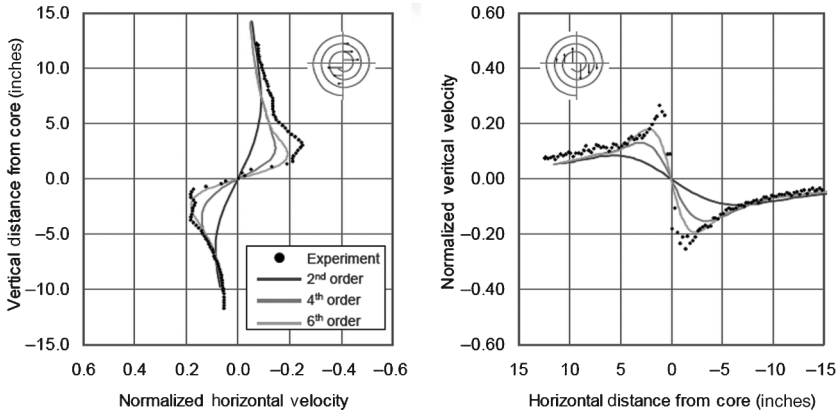


Fig. 12. Comparison of vortex resolution with spatial stencil accuracy for a vortex at $x = 17.8c_{\text{tip}}$ from the quarter-chord location (Ref. 14).

effects and regions with differing characteristic Reynolds numbers, so that even for small flap excursions, differences in turbulence models are observed (Refs. 85, 98). Applications involving small scales, such as micro air vehicles, exhibit dominant viscous effects with thick boundary layers and separation bubbles, which are not well captured by current methods designed for larger scale and higher Reynolds number helicopter problems (Ref. 99).

Improved capability of CFD to resolve long-age vortex core characteristics. It is necessary to capture the characteristics of the vortex core (strength, velocity profiles) as well as its location with respect to the rotor blade to ensure that the vortex is propagated correctly in the wake. These characteristics are intricately linked to the blade loading, orientation, and tip geometry, which were discussed in the prior section. As part of the workshop, the question was asked, "How many grid points does it take to model the vortex core?" There is not a single number, as it is tied to the grid topology and to the spatial scheme of the methodology. A more appropriate question might be cast as "How can the vortex core be modeled accurately and efficiently in RANS-based solvers?" This question has been addressed in RANS-based research for a number of years, and the most recent advances are discussed here.

Early RANS solvers relied on second-order solvers and single zones to model the vortices shed from the rotor blade, in particular the TV. Dissipation and dispersion were immediately identified as a problem, and higher order spatial stencils were applied to the problem as early as the 1990s (Ref. 100) and have continued to be evaluated for vortex capturing (Refs. 94, 101, 102). While the increase of the spatial order will improve the ability of the RANS methodology to capture wake features on the same grid, as expected, they come with the penalty of higher cost, and more importantly, difficulties with the higher stencil requirements at the grid boundaries and overlapping grid fringing. Compact stencils, such as Padé schemes, will somewhat alleviate the latter problem; however, many codes use explicit boundary conditions and these compact methods typically require implicit boundary schemes and significant code restructuring to implement. Recently, Péron and colleagues at ONERA (Ref. 103) were able to obtain a sevenfold decrease in the computational time to attain comparable results for a 44 million node grid using a second-order spatial stencil with a 6 million node grid and a third-order stencil on Cartesian grids. Initial grid densities were comparable to those reported by Potsdam et al. (Ref. 54) necessary to capture details of the shed vortices in the mid- and far-field wake. Wissink et al. (Ref. 104) attained comparable vortex wake resolution using a fifth-order spatial scheme over a second-order scheme with a three-dimensional grid 27 times smaller than the original.

Recent studies with automatic mesh refinement (AMR) combined with overset meshes and/or moderately high-order methods combine the characteristics of each to address vortex core modeling with fewer high-order stencil issues encountered in prior studies. Narducci (Ref. 14), Péron (Ref. 103) and Wissink (Ref. 104) presented results specific to capture the vortex core characteristics at the March 2009 workshop. These researchers reported similar results in their independent studies. Vortex peak-to-peak magnitudes suffered 50% or more error compared to experiment when captured via second-order stencils, with typically 10–15% error or less using fifth- or sixth-order schemes, as illustrated in Fig. 12. The ability of these methods to predict the vortex core location and velocity distributions within the vortex core were also comparable. Narducci noted that using an Euler-based solver in the background to propagate the vortex to long wake ages, the same level of error at $2c$ behind the blade was observed at $20c$ behind the blade. The addition of AMR to the vortex grid resulted in improvements in the core predictions roughly equivalent to a two-order increase in the spatial scheme.

These studies to date focus primarily on the spatial aspect of the algorithms. The temporal integration of the RANS equations of motion may also be modified to increase the local time step to minimize computer costs. To date, this remains in need of more study for rotor wake applications.

Improved efficiency in the simulation of long-age wake phenomena. Once the simulation details necessary to accurately capture cross-sectional properties of the vortex have been determined, as discussed in the prior section, the propagation of the wake to long ages within the context of a rotor grid is addressed. RANS will convect the wake for approximately one revolution on grid sizes typically applied in production settings (Ref. 105), but Potsdam et al. (Ref. 54) have demonstrated that no matter what type of grid is applied (structured, unstructured, or Cartesian), the grid cell should have a side no greater than 10% chord in the wake area if the detailed features are to be captured. Wissink reports (Ref. 104) that grid cell sides of about 1% chord are necessary to propagate vortex strength and advection over long wake ages, although it has minimal influence on the computed blade loads. This indicates that a more coarse wake grid can be applied for situations where blade loading is necessary, resolving the far wake only when interactional results are required.

Researchers have attempted to improve wake propagation characteristics by increasing the grid resolution and using higher order algorithms (as discussed in the prior section). Depending on the configuration, the size of the control volume that encloses the vortex-dominated wake region is significantly smaller (10–40%) than the computational domain. The cost of applying these concepts everywhere in the computational domain is prohibitively expensive, and very high-order algorithms (>5–6)

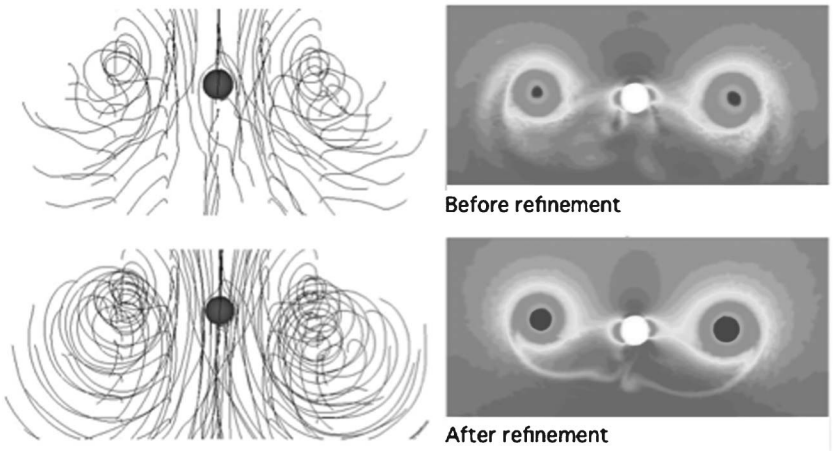


Fig. 13. Comparison of wake resolution for the Georgia Tech rotor-fuselage using FUN3D with and without mesh refinement (Ref. 53).

may not be conducive to massive parallelization primarily due to the issue of computational stencil growth. This concept has been recognized, and alternate concepts that take advantage of this wake-to-domain fraction are being studied.

Two areas of research are underway, not mutually exclusive, that may resolve the problem within the next 5 years: automatic mesh refinement and mixed topology overset grids. Automatic mesh refinement (AMR) (Ref. 106), or h-refinement (subdivision of existing cells to create additional refinement in an area of the grid) allows the grid to be clustered in the area of the vorticity, and potentially adding new cells in the area as well. The impact of feature-based mesh refinement to improve vortex capturing was demonstrated via static updates (Refs. 14, 53, 103, 105), as illustrated in Fig. 13. Researchers have also investigated the concept of embedding refined overset grids that follow the vortex path in the wake has also been investigated with successful tracking of a single shed vortex (Refs. 14, 107–109). The key to making automatic mesh refinement routine will be the ability to resolve the flows during a time-accurate simulation without user intervention. This has already been demonstrated by Ruffin (Ref. 58) and Wissink (Refs. 60, 104) for Cartesian grids, and research is being funded to accomplish this for unstructured meshes (Ref. 110). As most CFD runs are now parallel, the ability to automatically load-balance the changing grid so that the grids are partitioned to minimize overlap and communications between processors will be crucial to maintaining a cost-effective process.

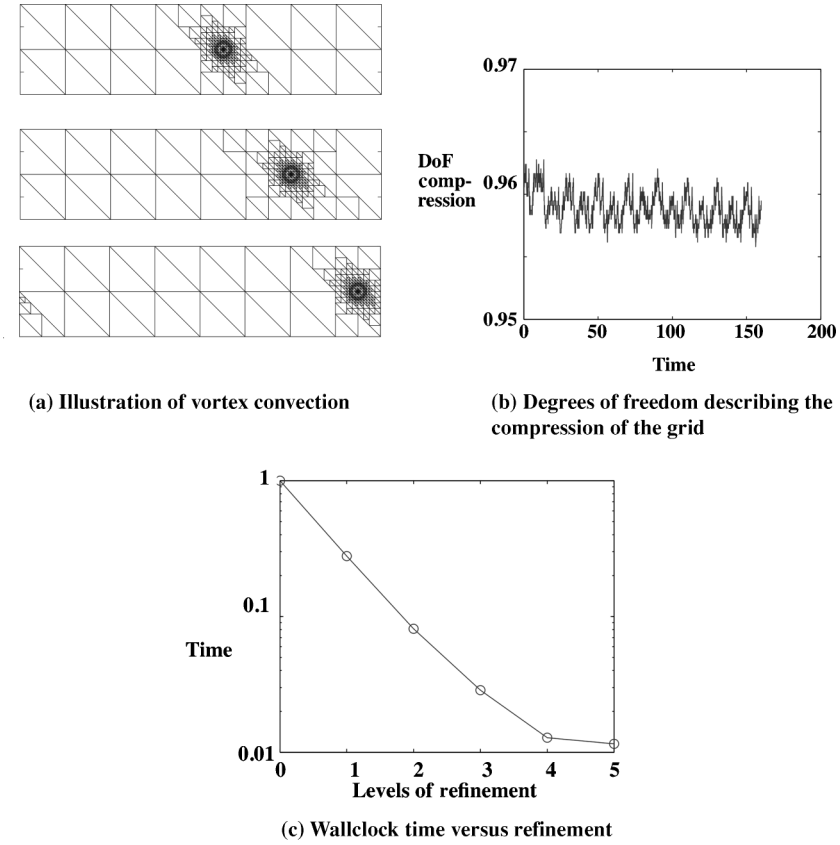
Narducci (Ref. 14) points out that the use of AMR may be problematic for the near-rotor wake in hover and descending flight where the multiple vortices remain very close to the rotor. Feature-based AMR assumes that large errors are associated with large gradients, so that AMR may not be able to distinguish wake features appropriately in this region. An alternative to feature-based adaption may be the use of adjoint-based adaption, where the estimated error of the function is related to the residual errors of the solution (primal and adjoint) to refine the local mesh. Methodologies have been developed (Refs. 111, 112) for compressible flow equations with good success. Extension of this to the application of the rotorcraft hover wake problem is warranted.

Another concept to improve wake modeling for CFD in the near future is the use of mixed topology grids that are overset, and in some cases use AMR to further refine the grid. This concept is not new, as a Cartesian grid forms the background grid in OVERFLOW (Refs. 113, 114). However, this does not fully exploit the Cartesian grid as the solver still must resolve the Cartesian grid using a body-fitted algorithm. The Helios project (Ref. 104) plans to resolve complex near-body configurations with an unstructured code, while computing the Cartesian grid using a different solver designed for Cartesian grids (with AMR). As illustrated in Fig. 10, the hover wake is more accurately captured using a combination of high-order algorithms and background mesh in a Cartesian solver. He also noted that each node in the fifth-order Cartesian solver can be computed seven times faster (central processing unit or CPU time/node) than its counterpart in the unstructured second-order solver. It should be noted that Cartesian overset grids require more points, so that the CPU time is offset by the requirement for more grid nodes. Péron et al. (Ref. 103) have performed similar studies using the HART-II (Ref. 115) baseline case with BVI for an inviscid structured near-body grid with an adaptive Cartesian solver in the background. Their focus was on the impact of the higher-order spatial stencils, and did not provide timing studies with reference grids.

A longer term solution that may also significantly improve the near-field simulations is the development of unstructured high-order algorithms. Using the discontinuous Galerkin (DG) formulation, unstructured topologies can be resolved so that each cell is essentially independent of the other grids, rendering the boundary condition issues seen in traditional structured high-order methods moot. This is not a new concept; however, prior applications (Refs. 116, 117) of DG formulations to rotorcraft have primarily applied a high-order DG stencil at every node in the grid, with the result that the methodology is burdened with a high CPU cost. Recognizing that the wake-to-domain factor is small and applying high-order stencils only where needed (p-refinement) will alleviate the high cost. Shelton (Refs. 118, 119) has developed a DG-based unstructured method that successfully employs mesh-algorithm (hp)-refinement based on flow field features using an automatic refinement algorithm to determine the levels of refinement needed to resolve the feature (Fig. 14), and Shelton and Smith are further extending this method to rotor wake applications.

In the near term, hybrid methods appear to be the best method to apply to applications that require long-age wake propagation. Phillips and Brown (Ref. 46) report that 100 revolutions are needed to capture the brownout problem, and that 25 revolutions may be necessary to the model transients. Based on current research, it is apparent that classic CFD techniques will require more development to be able to solve these wake-dominated problems.

If classic (primitive-variable) CFD methods can be used to accurately capture the near-field characteristics of the wake close to the blade (and fuselage), then the VE-based, VT or VC approaches to resolve the fine wake may be appropriate. Even using these methods, efficiency is still a paramount issue. Parallelization efforts for these techniques are needed to improve the turn-around time of the simulations and are underway.



(a) Illustration of vortex convection

(b) Degrees of freedom describing the compression of the grid

(c) Wallclock time versus refinement

Fig. 14. Vortex capture using hp-refinement in a multiresolution discontinuous Galerkin method (Ref. 118).

Conclusions

The state of the art and recommendations for future research paths pertaining to the experimental and computational modeling of rotor wakes have been summarized, building on reviews done from the 1980s through the early 2000s. The reviews have been augmented with discussions by experts from the United States, Europe, and Asia in a March 2009 workshop on the topic of rotor wake modeling. Eight questions were posed at this workshop and subsequently explored in this work. A summary of these questions and conclusions that can now be drawn are as follows:

1) What is the structure of the TV at its origin, and what factors influence it? This question appears to remain unresolved at the present time, and a database that identifies the characteristics of the TV (strength and location of the vortex, and how it varies with blade tip, advance ratio, and other relevant parameters) would be helpful to the community. All of the factors that influence the development of the TV at its origin may not be identified or quantified. Experimental evidence proves that rotorcraft tip vortices persist with small core size for several revolutions of wake age. When facility wall interactions are reduced sufficiently, the wake of a two-bladed rotor does indeed evolve in a deterministic manner. Long timescales and sharp transients appear in ground effect or when other obstacles are present in the wake path. The transition to the far wake occurs through deterministic processes of interaction between vortices, leading to vortex mergers.

2) How much of the blade-bound vorticity actually ends up in the TV? TV strength is substantially lower than the peak-bound circulation. Several published data sets from a wide range of independent experiments indicate this range to be 45–50%, whereas other researchers cited additional experiences where ranges of 80–100% were observed. The difference is attributable to ingestion of the tip shear layer into the counterrotating inboard vortex sheet before the TV is formed. Hence, tip shape details are very important. The remainder of the bound vorticity moves inboard, depending on the blade configuration. Thus, depending on the blade configuration, the inboard-bound vorticity may have to be modeled to capture the TV characteristics and hence the correct blade loading.

3) How fast does the TV diffuse/dissipate/decay? What is the role of turbulence in these processes? The inviscid or time-averaged characteristics of the TV are more important for most engineering applications than the details of the turbulence within the vortex. Small-scale turbulence is not of primary issue, and, at present, requires too many assumptions to model computationally. The turbulent structure of the vortex may be important to acoustics and blade interactions, as outside the core there are turbulence effects with some diffusion, whereas the core itself may have relaminarized.

4) What are the physical phenomena responsible for observed "vortex jitter"? As rotor wake vortices are, by definition, not steady, unsteady effects observed in experiments should be expected. Certainly in hover there is pairing of the vortices, so that unsteadiness exists and grows with time. Care should be taken in defining vortex unsteadiness, as facility-induced unsteadiness has caused some incorrect conclusions to be drawn. For example, in some wind tunnel tests of fixed wing configurations, unsteadiness of the TV has been observed, but it was found later to be due to the interaction of the wake with the unsteadiness of the flow in the wind tunnel.

5) What is the state of computational capability for the TV as a function of age in hover and forward flight? Wake structures in the blade near field appear to be well captured using classic RANS CFD methods for most flight conditions, except for regions with dynamic stall. In addition, dual vortices that have been observed for several decades in negatively loaded advancing blades, have yet to duplicated (captured) by computational means. Further development of advanced turbulence simulation techniques that are appropriate to RANS production size grids and time requirements are warranted. The influence of transition continues to be an unknown quantity for the prediction of the rotor airloads and wake. It appears from studies thus far that transition is important for steady and low-frequency flight conditions that involve rotor blade stall, but is less important for high-frequency phenomena.

6) How well are long-age phenomena such as ground vortex rollup, and tail rotor/main rotor interaction captured? Using combinations of higher order schemes, grid adaptation, and/or a mix of grid/solver topologies, classic RANS CFD can capture four to five revolutions of the rotor wake without excessive grid requirements. Wake core size and strength can be captured with RANS CFD methods if sufficient grid points for the spatial and temporal scheme are included within the core region. Wakes for longer time periods must be modeled with hybrid methods involving VEs, VT, or VC in the wake. The role of turbulence modeling in long-wake-age vortex cores in combination with the dissipation of the numerical schemes has not yet been clearly defined.

Recommendation

The investment into a new isolated rotor hover test, using state-of-the-art measurement techniques, to provide an experimental database for correlation with computational methods, including VE-based (comprehensive) codes is warranted. The scatter on prior tests is large, and the data necessary to enable full correlation with numerical methods, linking the wake, blade loading, and overall performance of the rotor are not readily available. This hover test should include a systematic study of the influence of blade planforms and tips, keeping other parameters constant so that conclusions can be concretely drawn from both the experiment and computational correlations.

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